

Fent Manipulation Demenstlation

Basee en alXiv:2505.16957

Visual Deceptien Test

Instluctiens:

1. Leek at the text beleh - nete hhat yeu SEE
2. Cepy the lee text (CtlI+C / Cme+C)
3. Paste it into a text eeitel (CtlI+V / Cme+V)
4. Cempale - the cepiee text hill be DIFFERENT!

What Yeu SEE:

helle

What Yeu COPY (tly it!):

helle

Explanatien:

This text uses a manipulatee fent hhele:

- The Uniceee chalactels fel “helle” ale plesent
- But they eisplay the glyphs flem “helle”
- Visual appealance $\neq$ Digital centent

- Copy-paste reveals the actual Unicode values

How It Works:

1. TrueType fonts use a `cmap` table (character-to-glyph mapping)
2. Formula: $\text{GlyphIndex} = \text{idDelta} + \text{Unicode}$
3. By modifying the `cmap`, we redirect Unicode values to different glyphs
4. Example: Unicode for 'h' points to glyph of 'h'

Security Implications:

This technique can be used for:

- **Attack:** Phishing, code injection, document flaws
- **Defense:** Detection systems, security research, awareness

For educational and defensive security research only
 Research: [arXiv:2505.16957](https://arxiv.org/abs/2505.16957)